LTIMB10 - 26.11.2025 - D2 - DATA MODELING ‘&’ DIMENSIONAL MODELLING

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1. Dimensional Modeling (Star & Snowflake Schema)

Dimensional modeling is a method used in Data Warehousing to organize data for **fast reporting**.

Think of it like arranging a supermarket:

- **Fact tables = Billing counters**
- **Dimension tables = Product racks, Customer details, Date info etc.**

This structure helps analysts find answers quickly.

Star Schema (Most Common)

Shape looks like a ⭐ because one fact table is surrounded by dimension tables.

Example (Retail Industry)

Fact Table: `Fact_Sales`

Dimensions: `Dim_Customer`, `Dim_Product`, `Dim_Date`, `Dim_Store`

```
          Dim_Customer
               |
Dim_Date — Fact_Sales — Dim_Product
               |
            Dim_Store
```

Why is Star Schema used?

- Fast queries
- Simple to understand
- Best for dashboards (Power BI, Tableau)

❄️ Snowflake Schema

Snowflake = more normalization → dimensions break into mini-dimensions.

Example:

`Dim_Product` → Product + Category + Subcategory

Dim_Category ← Dim_Product → Fact_Sales → Dim_Date

When used?

- When dimensions have hierarchical data
- When storage cost must be minimized
- When dimensions require frequent updates

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2. Fact & Dimension Tables

Fact Table

Stores measurable, numeric business events.

Examples:

- Sales amount
- Quantity sold
- Revenue
- Clicks
- Page views

Characteristics

- Contains **foreign keys** to dimensions
- Contains **metrics/measures**
- Usually **large** (billions of rows)

Dimension Table

Describes business entities.

Examples:

- Customer details
- Product details
- Dates
- Stores/Regions

Characteristics

- Text attributes
- Easier to read
- Connected to facts

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3. Slowly Changing Dimensions (SCD)

Dimensions change over time. SCD techniques preserve OR overwrite history.

SCD Type 1 – Overwrite (No History)

Old value replaced by new value.

Example:

Customer changes phone number → update existing row.

Use Cases:

- Data correction
- Non-critical history (contact details)

Old Phone → New Phone (overwritten)

SCD Type 2 – Add New Row (Full History)

A new row is added with **start_date**, **end_date**, **is_current flag**.

Example:

Customer changes address

→ We keep old address + new address.

Cust_ID | Address | Start | End | Current
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101 | Chennai | 2023 | 2024 | N
101 | Bangalore | 2024 | Null | Y

Use cases:

- Sales history
- Customer behavior analysis
- Banking, insurance

SCD Type 3 – Store Limited History

Keeps only **1 previous value**.

Example:

current_city | previous_city

Use case:

- Rarely used
- For small changes where only 1-level history needed

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4. Granularity

Granularity = Level of detail stored in Fact table.

Examples:

Granularity	Example
High	Each item sold per transaction
Medium	Total sales per day
Low	Total sales per month

Industry Example:

E-commerce usually stores order-level facts

—> High granularity

Banking: daily balance facts

—> Medium granularity

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5. Surrogate Keys

A surrogate key is an **artificial key** (integer) used in dimension tables.

Example:

customer_sk INT AUTOINCREMENT

Why not use natural keys?

- Natural keys change
- Not system-generated
- Hard to maintain in SCD

Surrogate keys = Stable + unique + fast joins

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6. Normalization vs Denormalization

Topic	Normalized (OLTP)	Denormalized (OLAP/DWH)
Purpose	Store data efficiently	Fast reporting
Structure	Many tables, many joins	Fewer tables, fewer joins
Example	Banking System	Data Warehouse
Benefit	Less redundancy	Fast queries
Drawback	Slow reporting	Large tables

Industry Rule:

→ OLTP: Normalize

→ OLAP: Denormalize (Star Schema)